



شبكات الحاسب

9:00-11:00

10/9/2023

د/ابرام كمال وليم

Faculty of Computers & Information, Assiut University

Course code: IT351

3rd Level

Final Exam

Duration: 2 hours

Total Marks: 50

Reviewer: (Add the name)

This exam for the following program(s) : (Add the program's name(s))

INTRODUCTION TO CANDIDATES:

- 1- This assessment paper contains of **twenty (25) Multiple Choice Questions (MCQ)**. Each question is worth 2 marks and the maximum possible mark is **50**.
- 2- This is a **CLOSED BOOK** assessment.
- 3- Calculators are allowed.
- 4- You are **NOT** allowed to communicate with each other or search online for answers.

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* رقم الجلوس

162019089

3

* المستوى

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ الرابع 2013
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016
- ☐ الرابع 2017
- ☐ الرابع 2018
- ☒ الرابع 2019

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* البرنامج

- ☒ عام
- ☐ بايو

- ☐ هندسة
- ☐ ذكاء صناعي

5

* رقم المعمل

- ☐ ج٠
- ☐ د٠
- ☐ ه٠
- ☐ اب
- ☐ اج
- ☐ اد
- ☐ اه
- ☐ أ٢
- ☐ ب٢
- ☐ ج٢
- ☐ د٢
- ☐ ه٢
- ☐ أ٣
- ☐ ب٣
- ☐ ج٣

☐ د٣

☒ ه٣

☐ أ٤

☐ ب٤

☐ ج٤

☐ د٤

☐ ه٤

6

* رقم الكمبيوتر

16

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

PYLM

8

Which of the following is a valid subnet mask? (2 Points)

☒ 0.0.0.0

☐ 127.0.0.1

☐ 255.0.0.0

☐ 255.244.0.0

☐ None of the above

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A browser requests to download index.html from a Web server. The HTTP response header received by the browser is shown below.

Which of the following statements is TRUE?

(2 Points)

```
HTTP/1.1 200 OK
Accept-Ranges: bytes
Content-Type: text/html; charset=UTF-8
Date: Sun, 20 Sep 2020 12:30:48 GMT
Expires: Sun, 27 Sep 2020 12:30:48 GMT
Last-Modified: Thu, 17 Oct 2019 07:18:26 GMT
Set-Cookie:      PHPSESSID=6puvcfqiu7f365rf7vc6
path=/
Content-Length: 1256
```

- ☐ In the HTTP request, browser has requested for a non-persistent connection.
- ☒ The Web server uses a cookie to keep the state information of the client.
- ☐ Suppose TCP header is 20 bytes, the length of the TCP segment containing the HTTP response is 1276 bytes.
- ☐ The index.html file received by the browser may have been corrupted during transmission.
- ☐ None of the above

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Which of the following is a public IP address? (2 Points)

- ☐ 10.10.10.10
- ☐ 172.31.31.31
- ☐ 172.17.17.17
- ☒ 192.169.10.10
- ☐ None of the above

11

A Web server stores a webpage that comprises a base HTML file and 2 images referenced by the base HTML file. The HTML file is 100 bytes and each image is 200 bytes. A client is connected to the Web server through a direct link of 1 Mbps. Propagation delay between the Web server and the client is 50 milliseconds.

The client downloads the webpage using persistent HTTP but without pipelining (i.e. the next HTTP request is sent after the response for the previous HTTP request is received). Assume HTTP header, TCP header and ACK packets are of negligible size, transmission channel is perfectly reliable, time to establish and close TCP connection can be ignored. How long (in milliseconds) does it take for the client to download the entire webpage from the Web server?

(2 Points)

- ☐ 204
- ☐ 200
- ☒ 304
- ☐ 404
- ☐ None of the above

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Which of the following statements is true?

(2 Points)

- ☐ Both the Internet and traditional telephone networks use packet-switching.
- ☐ Performance metrics such as end-to-end delay and throughput can be guaranteed in the Internet, if TCP is chosen as the transport layer protocol.
- ☒ The Internet uses packet switching and traditional telephone networks uses circuit switching.
- ☐ In the Internet, packets from the same source always take the same path to reach destination.
- ☐ None of the above

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Question Consider the following Python code snippet.

Suppose no exception is raised when the above code snippet is executed, which of the following statements is TRUE? (2 Points)

```
...  
s1.bind(('', 8080))  
s2, addr = s1.accept()  
...
```

- ☐ s1 is a UDP socket
- ☐ Server uses s1 to transmit application data to client
- ☐ s2 listens to a random port number assigned by operation system
- ☒ s2 listens to port 8080
- ☐ None of the above

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What is the checksum (1's complement of the sum) of the following 3 bytes?

11110100

10010011

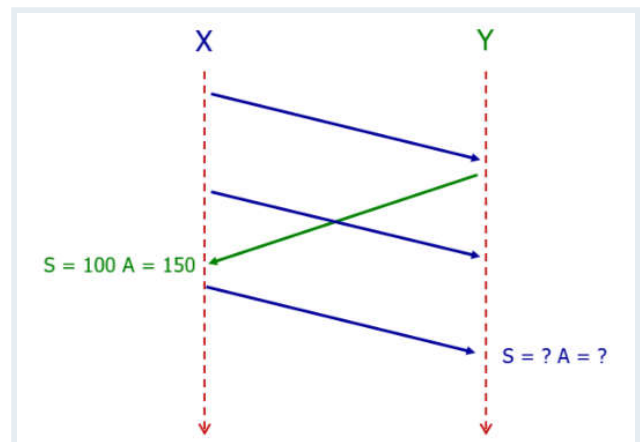
10011001

(2 Points)

- ☐ 111011101
- ☐ 101101101
- ☒ 11011101
- ☐ 11001101
- ☐ 11011100
- ☐ None of the above

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The following diagram shows two hosts X and Y communicating over a channel using TCP. X and Y are sending data to each other. Each TCP segment contains 100 bytes of data. None of the segments shown in the figure are retransmitted, out-of-order or corrupted packets. The first segment received by X has the sequence number 100 and ACK number 150.



What would be the ACK number (A) in the last segment received by Y?

(2 Points)

- ☐ 250

- ☒ 200

- ☐ 100
- ☒ 150
- ☐ 50
- ☐ None of the above

16

Which of the following commands would NOT cause a DNS query to be generated? (2 Points)

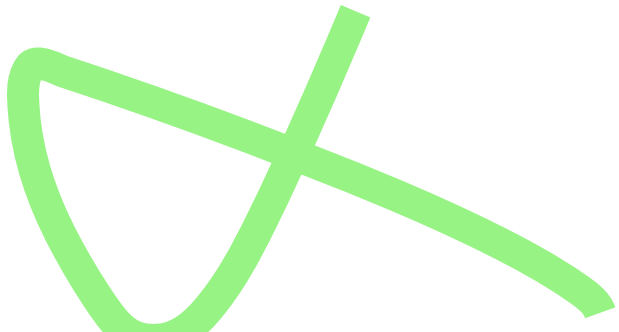
- ☐ dig a www.aun.edu.eg
- ☐ telnet www.example.org
- ☐ curl -I www.aun.edu.eg/fci/
- ☒ ping localhost
- ☐ nslookup www.google.com

17

There are many nodes in a shared medium network and most nodes are likely to transmit frequently. Which of the following multiple access protocol(s) is (are) suitable?

(2 Points)

- ☒ TDMA
- ☐ CSMA
- ☐ ALOHA



- ☒ Token passing
- ☒ FDMA
- ☐ All of the above
- ☐ None of the above

18

A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, x bits of the currently-being-transmitted packet have been transmitted on this outbound link and n other packets are waiting to be transmitted. Packets are transmitted continuously in order of arrival. Suppose all packets are L bits and the link rate is R bps.

What is the queuing delay?

(2 Points)

- ☐ nL / R
- ☐ L / R
- ☐ $n(L - x) / R$
- ☒ $(nL + (L - x)) / R$
- ☐ $nL + (L - x) / R$
- ☐ $(nL + (x - L)) / R$

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Which of the following layers doesn't appear in the Internet Protocol Stack?

(2 Points)

- ☐ Application layer

- ☐ Network layer
- ☐ Transport layer
- ☒ Session layer
- ☐ Physical layer

20

Which of the following statements is true about circuit switching?
(2 Points)

- ☐ Packet transmission can use the full link bandwidth
- ☐ The path between a source and a destination is not fixed
- ☒ Resources are pre-allocated regardless of demand
- ☐ None of the above
- ☐ All of the above

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Which of the following statements about DHCP is FALSE? (2 Points)

- ☐ DHCP is widely adopted in large networks where manual configuration is troublesome.
- ☒ DHCP server dynamically assigns an IP address to a host joining a network
- ☐ DHCP can only allocate private IP addresses
- ☐ DHCP server provides network configuration parameters such as subnet mask to a joining host
- ☐ DHCP messages are encapsulated in UDP segments

22

Which of the following statements about DNS is TRUE?

(2 Points)

- ☐ TLD servers store all the hostname to IP mappings of the Internet.
- ☐ Every Web server must have a canonical name.
- ☐ DNS server listens to TCP port 53.
- ☒ Local DNS server sometimes may provide out-of-date hostname-to-IP-address mapping.
- ☐ None of the above

23

Consider the generator, $G = 1001$, what is the CRC value R if data $D = 11000111010$? (2 Points)

- ☒ 110
- ☐ 011
- ☐ 101
- ☐ 001
- ☐ 111
- ☐ None of the above

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A TCP client sends a sequence of packets to a TCP server. Each packet is 1000

bytes

and the link rate is 2 Mbps. The propagation delay between the client and server is 6 ms. Which of following is the best timeout value chosen by the TCP client? (2 Points)

☐ 7 ms

☐ 9 ms

☐ 11 ms

☐ 14 ms

☒ 16 ms

☐ 18 ms

25

Consider a datagram network using 8-bit host addresses. Suppose a router uses longest prefix matching and has the following forwarding table.

What is the number of destination host addresses that is associated with interface 3?

(2 Points)

Prefix Match	Interface
10	3
1010	1
110	2
otherwise	0

☐ 40

☐ 58

☐ 48

☐ 50

☒ 128

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Suppose 5 data packets arrive at a TCP receiver. Each packet contains 10 bytes of application data. The sequence numbers of these 5 packets (shown in the order of arrival) are 50, 60, 90, 70 and 80 respectively. After receiving each packet, the TCP receiver sends an ACK packet. Suppose the first ACK packet has ACK number 60. What is the ACK number of the 3rd ACK packet?

(2 Points)

☐ 60

☐ 50

☐ 90

☒ 70

☐ 80

☐ None of the above

27

An IP address block 192.168.108/19 can be further divided into **X** subnets, each supporting a maximum of **Y** hosts. Which of the following is NOT a valid assignment? (2 Points)

☒ X = 4; Y = 2000

☐ X = 32; Y = 260

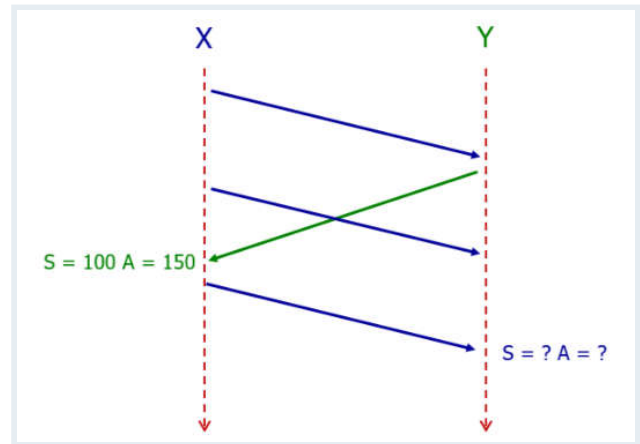
☐ X = 64; Y = 125

☐ X = 128; Y = 60

☐ X = 256; Y = 30

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The following diagram shows two hosts X and Y communicating over a channel using TCP. X and Y are sending data to each other. Each TCP segment contains 100 bytes of data. None of the segments shown in the figure are retransmitted, out-of-order or corrupted packets. The first segment received by X has the sequence number 100 and ACK number 150.



What would be the sequence number (S) in the last segment received by Y?
(2 Points)

☒ 250

☐ 200

☐ 100

☐ 150

☐ 50

☐ None of the above

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In Go-back-N protocol, what are the size of the respective sender and receiver buffers required for a window size of N?

(2 Points)

- ☐ 1; 1
- ☐ N - 1; 1
- ☒ N; 1
- ☐ N - 1; N - 1
- ☐ N; N

30

Which of the following is present in both HTTP request line and status line?

(2 Points)

- ☐ Request method
- ☐ Status code
- ☒ HTTP version number
- ☐ URL
- ☐ None of the above

31

Consider the transmission between a UDP sender and a UDP receiver. Which of the following will never happen? You may assume that the application riding on UDP doesn't implement any reliability mechanisms.

(2 Points)

- ☐ UDP receiver fails to receive any packet from UDP sender.
- ☒ UDP receiver receives duplicate packets from UDP sender.

- ☐ UDP receiver receives out-of-order packets from UDP sender.
- ☐ Option 5 UDP receiver receives corrupted packets from UDP sender but fails to detect bit errors.
- ☐ None of the above

32

A source and a destination are separated by 3×10^5 kilometers and are connected by a direct link of 2 Kbps. The propagation speed over the link is 2×10^8 meters/second. The source sends 100 packets to the destination using RDT 2.2. Each packet is of 2×10^3 bits long.

Suppose ACK packets are of negligible size and transmission channel is perfectly reliable. What is the throughput (in bps) of transmission?
(2 Points)

- ☒ 500
- ☐ 550
- ☐ 600
- ☐ 450
- ☐ 400
- ☐ None of the above



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